

Chat System Phase 2: Messenger Core Features

- ☐ 1. **Socket Service Extensions** (`src/app/services/socket-helper.ts`)
 - ☐ Add event endpoints: `getPrivateHistory` , `newPrivateMessage` , `privateHistory` , `newNotification` .
 - ☐ Support private messaging emitting logic.
 - ☐ Add Observables for private messages and notifications.
- ☐ 2. **Friends & Users API Integration** (`src/app/services/data.service.ts`)
 - ☐ Add API `GET /users/search?q=xxx` to search users.
 - ☐ Add API `GET /friends` and `GET /friends/pending` .
 - ☐ Add API `POST /friends/request/:id` and `PATCH /friends/accept/:id` .
- ☐ 3. **Search & Friend Requests UI** (`src/app/components/sidebar/sidebar.ts` or `Header`)
 - ☐ Implement search input capturing user queries.
 - ☐ Show search results with "Ajouter" button for creating a friend request.
 - ☐ Dynamic button state updates ("Envoyé").
- ☐ 4. **Notifications System** (`src/app/components/navbar/navbar.ts`)
 - ☐ Listen to `newNotification` via Socket.
 - ☐ Increment notification badge automatically.
 - ☐ Implement notifications dropdown showing Pending Requests & latest alerts.
 - ☐ Add "Acceptor" button inside pending friend requests in the dropdown.
- ☐ 5. **Sidebar Updates: Friends list vs Rooms**
(`src/app/components/sidebar/sidebar.ts`)
 - ☐ Fetch friends list upon component init.
 - ☐ Display friends alongside rooms (perhaps categorized or unified).
 - ☐ Click on friend sets active context to `Private Messaging` with `recipientId` .
- ☐ 6. **Chat Area: Private vs Group Mode** (`src/app/components/chat-area/chat-area.ts`)
 - ☐ Alter active context subscriptions to distinguish between selecting a `Room` or a `Friend` .
 - ☐ If Friend: emit `getPrivateHistory` and listen to `privateHistory` . Send messages using `{ recipientId: friend._id, content }` .
 - ☐ Combine history listener so both flows reuse the same unified display logic seamlessly.